

Image Segmentation using Joint Spatial-Intensity-Shape Features: Application to CT Lung Nodule Segmentation

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ABSTRACT

Automatic segmentation of medical images is a challenging problem due to the complexity and variability of human anatomy, poor contrast of the object being segmented, and noise resulting from the image acquisition process. This paper presents a novel feature-guided method for the segmentation of 3D medical lesions. The proposed algorithm combines 1) a volumetric shape feature (shape index) based on high-order partial derivatives; 2) mean shift clustering in a joint spatial-intensity-shape (JSIS) feature space; and 3) a modified expectation-maximization (MEM) algorithm on the mean shift mode map to merge the neighboring regions (modes). In such a scenario, the volumetric shape feature is integrated into the process of the segmentation algorithm. The joint spatial-intensity-shape features provide rich information for the segmentation of the anatomic structures or lesions (tumors). The proposed method has been evaluated on a clinical dataset of thoracic CT scans that contains 68 nodules. A volume overlap ratio between each segmented nodule and the ground truth annotation is calculated. Using the proposed method, the mean overlap ratio over all the nodules is 0.80. On visual inspection and using a quantitative evaluation, the experimental results demonstrate the potential of the proposed method. It can properly segment a variety of nodules including juxta-vascular and juxta-pleural nodules, which are challenging for conventional methods due to the high similarity of intensities between the nodules and their adjacent tissues. This approach could also be applied to lesion segmentation in other anatomies, such as polyps in the colon.

Keywords: Mean shift, Modified expectation-maximization (MEM), Pulmonary nodules, Shape index, Segmentation

1. INTRODUCTION

Lung nodule segmentation in CT images is a challenging medical imaging task. Nodules are often difficult to detect due to noise resulting from the image acquisition process, as well as the characteristics of the object itself and its neighborhood, particularly when the object has low contrast, a small size, or is located within an area of complicated anatomy [1]. For example, it becomes difficult to properly separate a nodule from adjoining tissues that have similar intensity characteristics, such as a blood vessel (juxta-vascular nodule) and the lung wall (juxta-pleural nodule).

Several lung nodule segmentation methods have been reported in literature [2-8]. An early method of nodule segmentation was described in [2] where dynamic programming was applied on each cross section of a nodule separately. This method relies on manual initialization and is a 2D approach. Other approaches based on deformable surfaces can be found in [3][4]. To increase robustness, several methods impose an *a priori* model; for example, the approaches described in [5] and [6] utilize shape models for spherical and ellipsoidal nodules respectively. This is also the case for the method introduced in [7], where the volumes of nodules were estimated using volume of an ellipsoid. However, such models are limiting in that not all types of lung nodules are accurately represented with these shapes. In [8], an adaptive sphericity-oriented contrast region growing method was used to segment nodule boundaries. The region growing is applied to the fuzzy connectivity map within a volumetric mask created by local adaptive segmentation. The method was tested on 815 pulmonary nodules and 84% of the segmentation results were accepted by radiologists.

In this paper, we have developed a novel feature-guided segmentation method that combines mean shift clustering and volumetric shape index to automatically segment lesions. The proposed algorithm has the following steps: it (1) calculates volumetric shape index (SI) at each voxel; (2) combines the SI with the intensity and the spatial position (x, y, z) to form a five-dimensional feature vector; (3) groups the five-dimensional feature vectors into clusters using the mean shift framework; (4) employs modified expectation-maximization algorithm (MEM) to merge the neighbouring

regions using the intensity mode map from the mean shift clustering. The joint spatial–intensity–shape (JSIS) feature provides rich information for the segmentation of the anatomic structures of interest, such as lung nodules. It can be seen that, unlike prior methods that compute features from an existing segmentation, our method integrates the features (e.g. intensity and shape) into the process of segmentation algorithm itself. Theoretically, by using the feature-guided segmentation, accurate object boundary delineation can be obtained. The experimental results on a CT lung nodule dataset demonstrate the performance of the proposed method. Most of the nodules are properly separated from the lung parenchyma or other attached tissues (e.g. vessel or lung wall). An accurate segmentation provides a solid base for detection, feature calculation, and classification in a CAD system.

The paper is organized as follows: Section 2 describes the main methodology of the proposed algorithm; Section 3 presents our experimental results; followed by a conclusion in Section 4.

2. METHODOLOGY

The mean shift framework provides a very effective methodology for feature space analysis. Initially developed by Fukunaga and Hostetler [9], mean shift has recently been exploited in low level computer vision tasks by Comaniciu and Meer [10]. It is a non-parametric estimator of the density gradient and does not require prior knowledge of the number of clusters, nor does it constrain the shape of the clusters. In this paper, a volumetric shape index feature is calculated at each voxel and combined with the intensity and the spatial position. This joint spatial-intensity-shape (JSIS) feature is clustered using the mean shift framework. A modified expectation-maximization algorithm (MEM) that considers a spatial constraint on the mean shift result (called the *mode map* in this paper) is then used to merge the neighbouring regions (modes). We demonstrate that by using this feature-guided segmentation, accurate nodule boundary delineation can be obtained.

Figure 1 gives an overview of our proposed method. In the following sections, each step is described in detail.

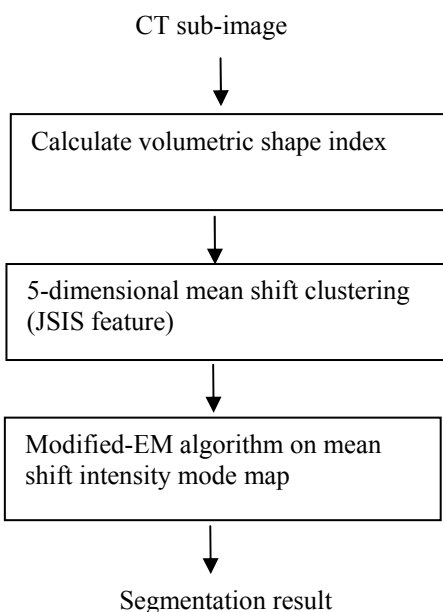


Fig.1 Flow chart of the proposed segmentation method.

2.1 Mean Shift Algorithm

Given n data points x_i , $i=1, \dots, n$ on a d -dimensional space R^d , the multivariate kernel density estimate obtained with

kernel $K(x)$ and window radius h is:

$$f(x) = \frac{1}{nh^d} \sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) \quad (1)$$

For radially symmetric kernels, it suffices to define the profile of the kernel $k(x)$ satisfying:

$$K(x) = c_{k,d} k(\|x\|^2) \quad (2)$$

where $c_{k,d}$ is a normalization constant which assures $K(x)$ integrates to 1. The modes of the density function are defined at the zeros of the gradient function, namely $\nabla f(x) = 0$.

The gradient of the density estimator (1) is:

$$\nabla f(x) = \frac{2c_{k,d}}{nh^{d+2}} \left[\sum_{i=1}^n g\left(\left\|\frac{x-x_i}{h}\right\|^2\right) \right] \cdot \left[\frac{\sum_{i=1}^n x_i g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}{\sum_{i=1}^n g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)} - x \right] \quad (3)$$

where $g(s) = -k'(s)$. The first term is proportional to the density estimate at x computed with kernel $G(x) = c_{g,d} g(\|x\|^2)$ and the second term:

$$m_h(x) = \frac{\sum_{i=1}^n x_i g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}{\sum_{i=1}^n g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)} - x \quad (4)$$

is the mean shift. The mean shift vector always points toward the direction of the maximum increase in the density.

The mean shift algorithm consists of the following steps:

- Select a start point x^0 ;
- Compute the mean shift vector $m_h(x^t)$;
- Move the feature point from x^t to x^{t+1} according the iterative equation: $x^{t+1} = x^t + m_h(x^t)$;
- If $\|x^t - x^{t+1}\| < \varepsilon$, then group all the points on the path from the start point x^0 to x^t into a cluster centered with x^{t+1} .

It is noted that the mean shift procedure is an adaptive gradient ascent method. The mean shift vector size depends on the gradient itself; when it is near local maxima, the step is small and refined. By acting in the joint spatial-intensity-shape domain, mean shift brings together many voxels with potentially significant intensity variations, into a few modes, which is called mode map.

2.2 Volumetric shape index: a 3D geometric feature

The volumetric shape index (SI) at voxel $p(x,y,z)$ can be defined as [11]:

$$SI(p)=\frac{1}{2}-\frac{1}{\pi}\arctan\frac{k_1(p)+k_2(p)}{k_1(p)-k_2(p)} \tag{5}$$

where $k_1(p)$ and $k_2(p)$ are the principal curvatures [11] at voxel p , which are defined as:

$$k_1(p)=H(p)+\sqrt{H^2(p)-K(p)},\; k_2(p)=H(p)-\sqrt{H^2(p)-K(p)}$$

where $K(p)$ and $H(p)$ are the Gaussian and mean curvatures.

The calculation of the Gaussian and mean curvatures are based on the first and second fundamental forms of differential geometry. A practical approach for computing the first and second fundamental forms at each voxel p is to use the smoothed first and second partial derivatives of the image with respect to x, y, z , as described in [12].

Shape index is a very useful feature. Theoretically, it represents the local shape feature at each voxel while being independent of the image intensity. Every distinct shape, except for the plane, corresponds to a unique shape index. For example, the shape index value is 1.00 indicates a sphere-like shape (e.g. lung nodule), and 0.75 indicates a cylinder-like shape (e.g. vessel). Based on the definition, volumetric shape index directly characterizes the topological shape of an iso-surface in the vicinity of each voxel without explicitly calculating the iso-surface. This feature provides rich information for automated segmentation of anatomical structures or tumors (lesions) in medical images, where the region of interest is within an area of complicated anatomy and image intensities of different shapes are very similar to each other (such as the adjoining lung nodule).

To illustrate the characteristics of the shape index, Fig. 2 (a) and (b) show an original lung image and its corresponding shape index map; (c) and (d) are highlighted shape index values on one nodule and one blood vessel, respectively. It is noted that different shape index values between the sphere-like nodule and the cylinder-like blood vessel at each voxel are obtained; the average of the shape index values for the nodule is higher than that of the blood vessel (e.g. the average shape index value around the nodule is about 0.9; while that of the vessel is about 0.75).

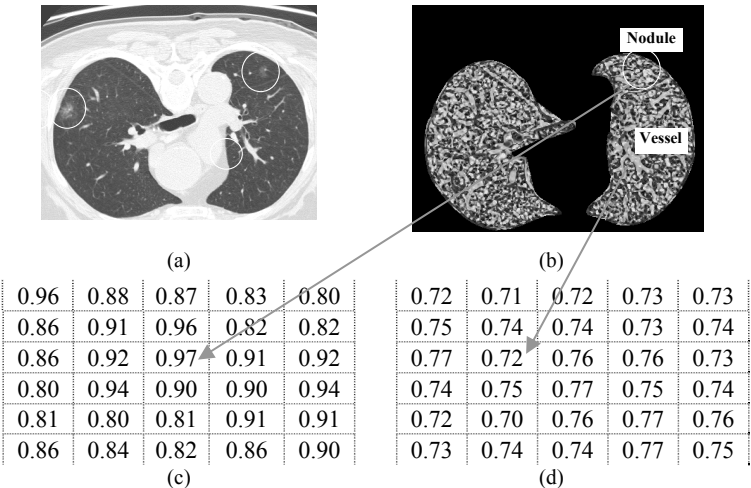


Fig. 2 An example of shape index map with nodules (a) Original lung image; (b) Shape index map; (c) Shape index values for sphere-like nodule; (d) Shape index values for cylinder-like blood vessel.

2.3 Combination of shape index feature into mean shift framework

For each voxel, 3D spatial location, intensity and volumetric shape index features are concatenated in the joint spatial-intensity-SI domain of dimension $d=5$. Given n data points x_i , $i=1, \dots, n$ on a 5-dimensional space R^5 , (where n is the total number of voxels in the image), the multivariate kernel in (2) is defined as the product of three radially symmetric kernels:

$$K_{h_s, h_r, h_{si}}(x) = c_{k,s} k\left(\left\|\frac{x^s}{h_s}\right\|^2\right) \cdot k\left(\left\|\frac{x^r}{h_r}\right\|^2\right) \cdot k\left(\left\|\frac{x^{si}}{h_{si}}\right\|^2\right) \quad (6)$$

where x^s is the spatial location, x^r is the intensity and x^{si} is the shape index feature; $k(x)$ is the common profile used in all the domains; h_s , h_r and h_{si} are the kernel window size for spatial, intensity and shape index kernel function, respectively. The Normal kernel is used in this paper, where $k(x) = (2\pi)^{-d/2} \exp\left(-\frac{1}{2}\|x\|^2\right)$.

By using the mean shift framework, the shape index feature can be combined with the intensity for object segmentation. Similar to Equation (4), the mean shift vector with three kernel windows (spatial, intensity and shape index) can then be calculated as:

$$m_{h_s, h_r, h_{si}}(x_i) = \frac{\sum_{i=1}^n x_i g\left(\left\|\frac{x^s}{h_s}\right\|^2\right) \cdot g\left(\left\|\frac{x^r}{h_r}\right\|^2\right) \cdot g\left(\left\|\frac{x^{si}}{h_{si}}\right\|^2\right)}{\sum_{i=1}^n g\left(\left\|\frac{x^s}{h_s}\right\|^2\right) \cdot g\left(\left\|\frac{x^r}{h_r}\right\|^2\right) \cdot g\left(\left\|\frac{x^{si}}{h_{si}}\right\|^2\right)} - x_i \quad (7)$$

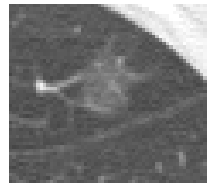
where $g(s) = -k'(s)$.

The joint spatial-intensity-shape index mean shift procedure estimates the modes (the densest regions) of the multivariate distribution underlying the feature space. The set of points that converge to the same mode is defined as the attraction basin. Based on the joint intensity, spatial and shape features, mean shift maps all the data samples to the local maxima of their corresponding attraction basin (mode map).

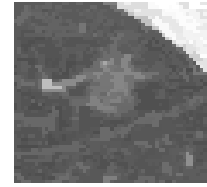
Figure 3 shows an example of one original CT sub-image containing one Ground-Grass Opacity (GGO) nodule attached to small vessels and its intensity mode map (b) by using 5-dimensional JSIS mean shift clustering. (c) and (d) are the corresponding values for the original CT image and its mode map at the same voxel in the nodule area. It is noted that the mode map can be seen as a “filtered” image and is less contaminated by outliers.

2.4 Modified expectation-maximization considering spatial information

The mode map (M_i) obtained by the above joint spatial-intensity-shape index mean shift algorithm expresses the local structure of the data in a given region of the feature space. The number of modes depends on the kernel window size and the data structure. Although this number is a large compression of the initial data, sometimes it is still larger than the target number of classes. Next, a mixture Gaussian model with expectation-maximization (EM) that considers



(a)



(b)

-786	-773	-655	-686	-708	-746	-741	-648	-689	-833
-692	-721	-705	-638	-648	-681	-582	-530	-745	-811
-659	-684	-721	-679	-586	-523	-538	-621	-718	-724
-730	-710	-726	-702	-571	-552	-585	-661	-719	-765
-764	-709	-704	-643	-595	-528	-696	-747	-738	-711
-691	-707	-666	-660	-664	-650	-674	-734	-739	-724
-670	-681	-636	-605	-625	-651	-596	-655	-701	-636
-738	-645	-673	-706	-716	-713	-723	-642	-625	-721
-672	-613	-724	-702	-672	-706	-643	-613	-707	-755
-573	-503	-610	-698	-739	-742	-627	-533	-679	-843
-688	-703	-728	-672	-753	-767	-621	-411	-621	-858

(c)

-778	-778	-608	-706	-706	-728	-728	-639	-728	-838
-704	-704	-706	-608	-608	-639	-639	-608	-728	-838
-706	-706	-706	-706	-608	-608	-608	-639	-728	-728
-704	-704	-704	-706	-608	-608	-608	-639	-728	-728
-704	-704	-704	-608	-608	-608	-608	-728	-728	-728
-704	-704	-704	-704	-608	-608	-608	-728	-728	-728
-608	-704	-704	-608	-608	-608	-608	-608	-608	-608
-754	-608	-608	-704	-728	-728	-728	-608	-608	-728
-608	-582	-681	-704	-704	-728	-608	-608	-728	-728
-582	-417	-582	-608	-754	-754	-608	-608	-728	-838
-754	-754	-754	-608	-754	-754	-608	-608	-608	-858

(d)

Fig. 3 An example of using 5-dimensional JSIS mean shift clustering on one GGO nodules (a) Original CT sub-image with one GGO; (b) Intensity mode map from mean shift clustering; Corresponding intensity values for original CT image (c) and its intensity mode map (d), at the same voxel in the nodule area.

the spatial information is applied to the mode map to merge neighboring modes. Based on the Bayesian probability theory, for each mode, the probability of the mode belonging to one class is defined as:

$$p(\varphi_l | M_i) = \frac{p_l(M_i | \varphi_l) \cdot p(\varphi_{l(i)})}{\sum_l p_l(M_i | \varphi_l) \cdot p(\varphi_{l(i)})} \quad l = 1, 2, \quad i = 1, 2, \dots, m \quad (8)$$

where m is the total number of the modes in the mean shift mode map, $p_l(M_i | \varphi_l)$ is l th Gaussian model with parameter $\varphi_l = (\mu_l, \sigma_l)$ (mean μ_l and standard deviation σ_l), and $p(\varphi_{l(i)})$ is a spatial prior probability. In this paper, two classes ($l=1,2$): object (nodule) and background, is considered.

It is noted that the spatial prior probability $p(\varphi_{l(i)})$ plays an important role in Equation (8). The spatial constraint can be imposed by a Markov Random Field (MRF) and Gibbs Random Field (MRF-GRF) [13] as follows:

$$p(\varphi_{l(i)}) \approx p(l_i | l_{N(i)}) = \frac{\exp\left[-\sum_c v_c(l_i)\right]}{\sum_{l_i} \exp\left[-\sum_c v_c(l_i)\right]} \quad (9)$$

where $N(i)$ is the neighbourhood of mode i . The potential function $v_c(l_i)$ associated with the clique c is defined as $v_c(l_i) = -\beta \cdot p(\varphi_l | M_i)$. In this paper, a second-order neighbourhood system and pair-wise interaction clique between neighbours are considered. β is a positive constant which controls the size of clustering.

The posterior probability $p(\varphi_l | M_i)$ in Equation 8 represents the probability that the given mode i belongs to one class $l \in [1, 2]$. Each mode is assigned to one Gaussian component that has the highest likelihood, namely, $l_i^* = \arg \max_l p(\varphi_l | M_i)$. By using the spatial constraint in Equation (9), the posterior probability $p(\varphi_l | M_i)$ not only depends on the mode intensity but also considers its local spatial information.

3. EXPERIMENTAL RESULTS

The proposed segmentation algorithm has been evaluated with a database of clinical chest CT scans, containing 68 nodules out of 50 scans, with a slice thickness ranging from 0.5mm to 2.0mm. The tube current ranged from 30mA to 250mA. The size of the nodules ranged between 5mm to 20mm in diameter. To produce ground truth data, each nodule boundary was manually delineated by an experienced radiologist.

For comparison, two experiments were carried out to demonstrate the performance of the proposed method. In the first experiment, the four dimensional spatial-intensity feature (without shape index) was used in the mean shift framework for the clustering, followed by a MEM algorithm on the mean shift intensity mode map. In the second experiment, the whole proposed method with the five dimensional spatial-intensity-shape feature was applied.

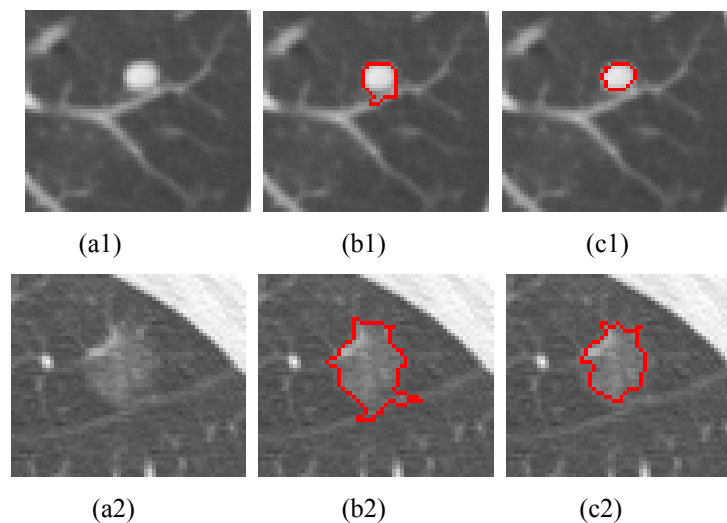


Fig.4 Two examples of nodule segmentation. 1st column: original sub-image; 2nd column: segmentation results based on mean shift without the shape feature; 3rd column: segmentation results based on the proposed method.

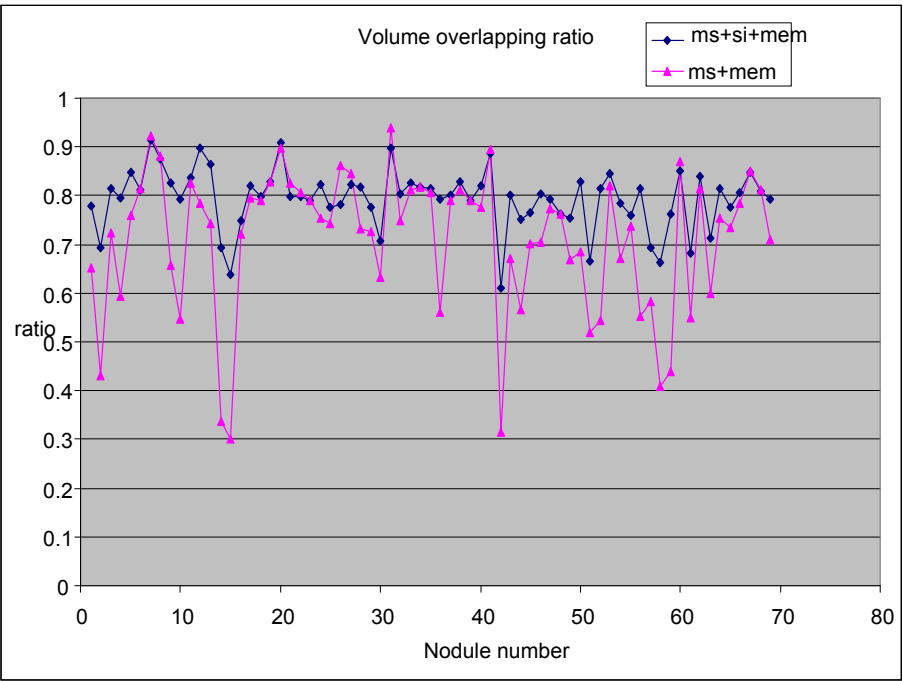


Fig.5 Volume overlap ratio based on the two different methods. The mean overlap ratio over the whole dataset for the proposed method is 0.80; and 0.71 for using MEM on the mean shift map without the shape feature.

Fig. 4 shows examples of nodule segmentation using the two different methods on two different nodules: 1) one solid nodule attached to vessel shown in Fig.4 (a1); 2) one pure GGO nodule adjoining to vessel shown in Fig. 4(a2). It can be seen that, by combining the shape index feature into mean shift framework, both of the nodules are properly delineated from the background lung parenchyma despite the presence of other non-target structures such as vessels. Especially, it is noted that, compared to solid nodules, GGO nodules usually have faint contrast and fuzzy margins as shown in Fig. 4(a2). However, by using the proposed method, accurate nodule boundary can be obtained as shown in Fig. 4(c2). This is because the shape index characterizes the local shape while being independent of the image intensity. By integrating this feature into intensity-based mean shift framework, the segmentation not only considers the intensity but also the object shape information.

To give the overall performance of the proposed algorithm, each nodule is segmented and compared with a ground truth manual segmentation. An overlap ratio between the segmented nodule and the ground truth annotation is calculated. Fig. 5 shows the overlap ratios based on the proposed method with and without the shape index feature, respectively. It is noted that, without the shape index combined into mean shift framework, the mean overlap ratio for the whole dataset is 0.71. However, the overlap ratio increases to 0.80 using the proposed method. This demonstrates the potential of the proposed method for accurate nodule segmentation.

4. CONCLUSION

This paper proposed a new method for lung nodule segmentation by integrating five-dimensional spatial-intensity-shape feature into the mean shift framework. The joint JSIS feature provides rich information for lesion segmentation. This is particularly useful when the region of interest is within an area of complicated anatomy for which the image lesion has similar intensities to its neighboring tissues but has a different shape (such as juxta-vascular nodules and juxta-pleural nodules). In addition, we applied a modified expectation-maximization algorithm to the intensity mode map from the mean shift clustering to merge the neighboring regions (modes). Both by visual inspection and quantitative measurement, results using a clinical dataset of thoracic CT scans confirm high performance of the proposed method. The method can successfully segment lesions adjacent to structures of similar intensity but different shape. The approach can also be applied to lesion segmentation in other anatomies, such as polyps in the colon.

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